Project Report: Census Data Income Classification and Segmentation Analysis

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Abstract

This study presents a machine learning analysis of the U.S. Census dataset, addressing two objectives: binary income classification and population segmentation. For classification, I employ **XGBoost** to predict whether individuals earn above or below \$50,000 annually, achieving an Area Under the Receiver Operating Curve (AUROC) of 0.95 on test data. To address the severe class imbalance, I apply **class based weighting** and evaluate performance across multiple decision thresholds, demonstrating how threshold selection can be tailored to specific business objectives such as **precision-first or recall-first** strategies. For segmentation, I apply **K-Means** clustering with **PCA** dimensionality reduction to identify eight distinct population segments. Cluster profiles are characterized using lift analysis for categorical features; and effect size metrics for numeric variables, revealing demographically interpretable groups including working spouses, dependent minors, retired seniors, and immigrant communities. Each segment is paired with **actionable marketing recommendations**.

1 Data Exploration

1.1 Exploratory Data Analysis

Visualization of all features reveal the distribution of numeric variables (e.g., age, capital gains) and categorical variables (e.g., education, marital status). These visualizations inform preprocessing decisions and help identify predictive relationships with income.

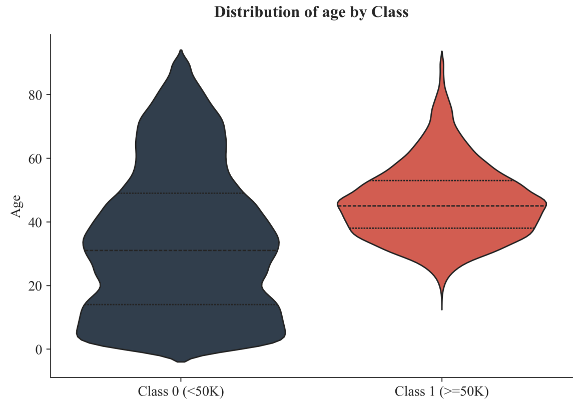
The target variable has strong class imbalance:

- Income <\$50K: 187,141 samples (93.8%)
- Income $\geq$\$50K: 12,382 samples (6.2%)
- Class ratio: **15.11 : 1**

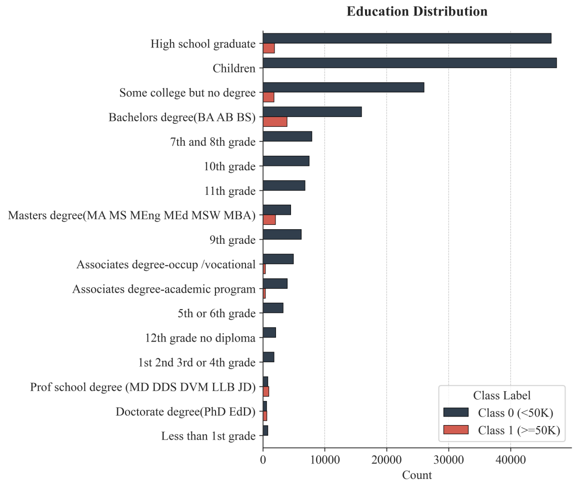
The negative class outnumbers the positive class 15-fold, necessitating class balancing techniques during model training to prevent bias toward predicting the majority class. Given below is one numeric and one categorical distribution for understanding the data (Figure 1). Plots for all features are available in the project repository in data plots folder.

1.2 Missing Values

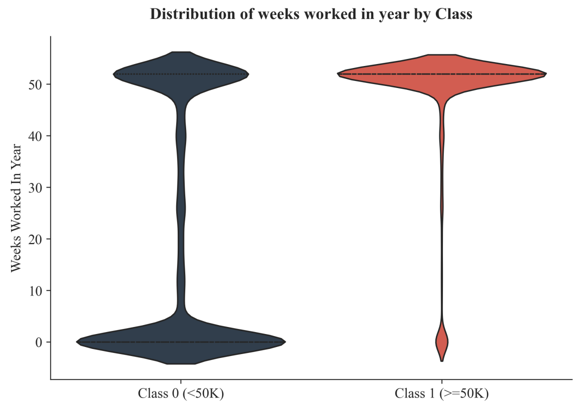
The Census dataset uses various string placeholders to denote missing values rather than explicit NaN entries. The missing values were counted based on these mappings to check what % of data is absent for each feature. Table 1 shows a glimpse of 10 features and their percentage of missing values.



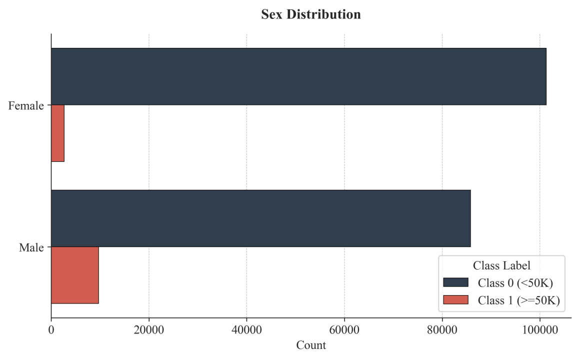
(a) Distribution of age by class



(b) Education distribution



(c) Distribution by weeks worked in a year



(d) Sex distribution

Figure 1: Figure 1 presents distribution plots for one numeric and categorical feature. Numeric variables are visualized using violin plots, which display the probability density, median, and interquartile range across income classes. Categorical variables are represented using bar plots.

1.3 Feature Importance Analysis

Feature importance analysis was conducted to identify the most predictive features and eliminate uninformative features. A **Random Forest** classifier was used to quantify feature importance through the following procedure:

- An **independent 80/20 train-test split** with stratified sampling is created specifically for this analysis to ensure unbiased importance estimates. This split differs from the final modeling pipeline to prevent data leakage during feature selection.
- A Random Forest classifier with 300 trees was trained on the training data. Random Forest was selected for importance calculation due to its built in **feature ranking based on mean decrease in Gini impurity** across decision trees.

Features with importance below 1% were excluded from final modeling, as these contribute negligibly to predictive performance while increasing model complexity and the risk of overfitting. This threshold resulted in retention of the **top 24 features** that capture the primary predictive signal in the data. Table presents the top 5 and bottom 5 features by importance. The complete ranking is available in the project repository.

Table 1: Top 5 and bottom 5 features by importance.

Feature	**Importance (%)**	**Missing (%)**
education	8.79	0.00
age	8.14	0.00
weight	8.11	0.00
detailed occupation recode	7.11	50.46
dividends from stocks	6.86	0.00
reason for unemployment	0.15	96.96
live in this house 1 year ago	0.15	50.73
enroll in edu inst last wk	0.07	93.69
fill inc questionnaire for veteran's admin	0.03	99.01
family members under 18	0.01	72.29

1.4 Data Processing

Several categorical features contain many unique values, which would produce **sparse one-hot encoded representations**. To mitigate this, semantically related categories were consolidated into broader groupings. The complete mapping definitions are available in the project repository. Table 2 provides an example.

Table 2: Example category mapping for household relationship.

Original Categories	**Mapped Category**
Householder, Nonfamily householder	Householder
Secondary individual, RP of unrelated subfamily	Non-family individual
In group quarters	Group quarters
Spouse of householder, Spouse of RP, 6 more...	Spouse or partner
Child 18+ ever marr, Grandchild 18+, 15 more...	Child or grandchild
Other Rel 18+ ever marr, 8 more...	Other relative

- Target binarization: income $\geq$\$50K $\rightarrow$ 1, income <\$50K $\rightarrow$ 0
- Data partitioning: 70% train, 15% validation, 15% test (stratified)
- Categorical encoding: one-hot encoding
- Numeric scaling: Z-Score normalization (zero mean, unit variance)

2 Classification

2.1 Model Architecture

XGBoost (Extreme Gradient Boosting) is used as for classification due to its state-of-the-art performance on tabular datasets and robust handling of class imbalance. Comparative studies (e.g., Shwartz-Ziv & Armon, 2022) demonstrate that tree based ensemble methods consistently outperform deep neural networks on structured data. The model is trained by minimizing the following objective function:

$$\mathcal{L}(\phi) = \sum_i l(\hat{y}_i, y_i) + \sum_k \Omega(f_k) \tag{1}$$

where ϕ represents the model parameters (i.e., the ensemble of decision trees), l is the logistic loss measuring the discrepancy between predicted probabilities $\hat{y}_i$ and true labels y_i, and $\Omega(f_k)$ represents regularization terms that penalize model complexity to prevent overfitting. The **first term drives predictive accuracy, while the second term encourages simpler trees**, balancing model fit against generalization.

Table 3: XGBoost hyperparameter configuration.

Value	**Description**
6	Maximum tree depth
1	Minimum sum of instance weights in child nodes
0.05	Learning Rate
2000	Maximum boosting rounds
50	Patience for early stopping
15.11	Class weight ratio

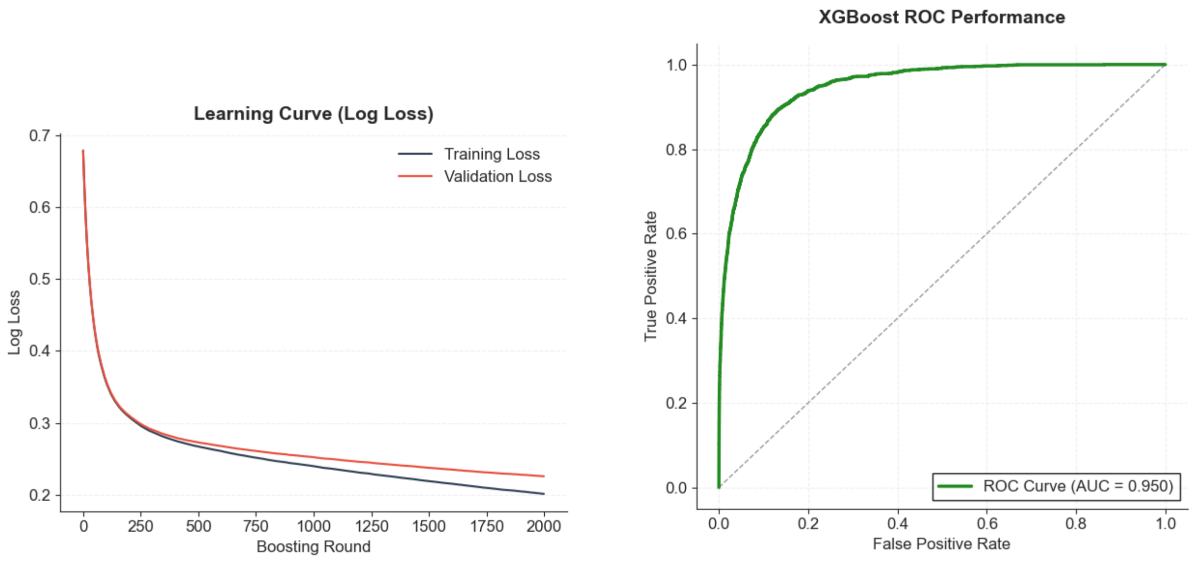
2.2 Evaluation

Model performance was assessed using multiple complementary metrics to provide a comprehensive view of classification quality

- Area Under Receiver Operating Curve (AUROC): Measures how well the model distinguishes between high income and low income individuals, with values ranging from 0.5 (random guessing) to 1.0 (perfect separation). This metric evaluates **overall model quality independent of any specific threshold**.
- Accuracy: Measures overall classification correctness at a **specified threshold**. While intuitive, accuracy can be **misleading under class imbalance**, as a naive classifier predicting all samples as the majority class would achieve 93.8% accuracy.
- Class specific Precision and Recall: For each class, precision measures the proportion of predicted positives that are truly positive , while recall measures the proportion of actual positives correctly identified. These metrics reveal **model performance separately** for high income (Class 1) and low income (Class 0) individuals.

Table 4: Classification performance across decision thresholds on the test set.

Threshold	**Accuracy**	**Class 1 (≥$50K)**		**Class 0 (<$50K)**
		Precision	Recall	Precision / Recall
0.40	90.40%	37.7%	**83.6%**	**98.8%** / 90.9%
0.60	94.42%	54.2%	64.8%	97.6% / 96.4%
0.80	95.14%	60.6%	61.8%	97.5% / 97.4%
0.95	**95.57%**	**82.4%**	36.3%	95.9% / **99.5%**



(a) Learning curve (log loss) (b) ROC curve (AUROC = 0.950)

Figure 2: Illustrates the model's learning dynamics and predictive performance. (a) Training and validation loss curves across boosting rounds, demonstrating convergence without overfitting. (b) ROC curve on the test set, achieving an AUROC of 0.95.

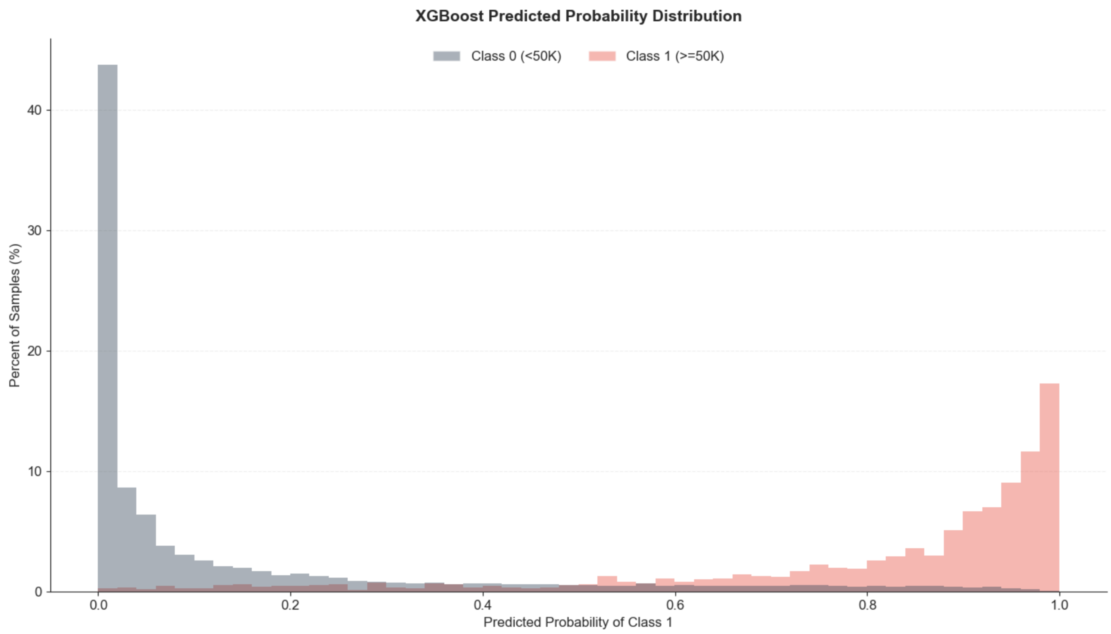


Figure 3: Displays the distribution of predicted probabilities separated by true class label, with gray representing low income individuals (Class 0) and red representing high income individuals (Class 1). An ideal classifier would concentrate Class 0 predictions near 0.0 and Class 1 predictions near 1.0. The observed distributions show separation for most samples, indicating strong discriminative performance, though some overlap in the mid range probabilities reflects the inherent difficulty of classifying ambiguous cases.

2.3 Analysis and Recommendation

2.3.1 High Precision First Strategy (threshold = 0.95)

When offering premium services, high confidence in predictions is essential. A threshold of 0.95 achieves 82.4% precision in which more than 4 out of 5 flagged customers truly earn $50K but recall drops to 36.3%, missing two thirds of actual high income individuals. This conservative approach minimizes costly false positives while accepting some opportunity loss.

2.3.2 High Recall First Strategy (threshold = 0.40)

For customer acquisition and marketing campaigns where contact costs are low, maximizing coverage becomes paramount. A threshold of 0.40 achieves 83.6% recall, capturing 5 out of 6 actual high income customers, but precision falls to 37.7%. This aggressive approach is economically rational when the opportunity cost of missing high value customers exceeds the cost of broader outreach.

3 Segmentation

3.1 Preprocessing

Unlike supervised classification, unsupervised clustering does not provide feature importance rankings to guide variable selection. Therefore, feature exclusion was based solely on data quality criteria: features with excessive missing values or minimal informational content were removed. Twelve features met these exclusion criteria, including year, state of previous residence, migration code, and veteran's administration questionnaire completion. The complete list is available in the project repository. All previous preprocessing steps categorical encoding, numeric scaling, missing value handling, etc **follow the same procedures as described previously. The one addition to this is Principal Component Analysis**.

3.1.1 Principal Component Analysis (PCA)

Principal Component Analysis is an unsupervised dimensionality reduction technique that transforms high dimensional data into a lower dimensional representation while preserving maximum variance. PCA identifies orthogonal directions (principal components) in the feature space along which data varies most, effectively compressing information by projecting data onto these key directions. It is applied to reduce feature space complexity. Components are selected to **preserve 99% of total variance.**

3.2 Model

K-Means is an unsupervised learning algorithm that partitions data into K distinct, non overlapping clusters by minimizing within cluster variance. The algorithm assumes clusters are spherical and roughly equal in size, making it well suited for identifying compact, well separated segments in high dimensional data.

The objective function to be minimized is:

$$J = \sum_{j=1}^{K} \sum_{i \in C_j} \|x_i - \mu_j\|^2 \tag{2}$$

where K is the number of clusters, C_j is the set of points in cluster j, and μ_j is the centroid of cluster j. The algorithm minimizes this objective iteratively through the following steps:

1. Initialize K cluster centroids randomly from the data points

2. Assign each data point to the nearest centroid based on Euclidean distance
3. Recalculate each centroid as the mean of its assigned points
4. Repeat steps 2 - 3 until convergence (centroid positions stabilize below a tolerance threshold) or the maximum iteration count is reached

To mitigate sensitivity to initialization, the algorithm executes 10 independent runs with different random seeds, selecting the solution with lowest inertia (within cluster sum of squares).

Table 5: K-Means hyperparameter configuration.

Value	**Description**
8	Number of clusters
10	Number of random initializations
Euclidean	Similarity measure

3.3 Evaluation

Cluster quality was assessed across $k \in \{3, \ldots, 11\}$ using three complementary metrics:

- **Silhouette Score**: Measures cohesion and separation, from -1 to 1 (higher is better)
- **Davies Bouldin Index**: Measures average cluster similarity, (lower is better)
- **Calinski Harabasz Index**: Ratio of between cluster to within cluster variance (higher is better)

Table 6: Cluster evaluation metrics across different values of k.

k	**Silhouette**	**Davies-Bouldin**	**Calinski-Harabasz**
3	0.186	**2.061**	**40293.2**
4	0.191	2.226	33154.3
5	0.199	2.271	29272.4
6	0.203	2.123	26504.0
7	0.200	2.087	24177.3
8	**0.204**	2.244	22178.7
9	0.158	2.292	20607.9
10	0.177	2.163	18760.1
11	0.153	2.217	17941.6

3.3.1 Cluster Size Analysis

The silhouette score peaks at $k = 8$ (0.204), indicating optimal cluster cohesion and separation. While $k = 6$ and $k = 7$ yield marginally better Davies-Bouldin scores (2.12 and 2.09 respectively), the differences are not very significant. Beyond statistical considerations, $k = 8$ aligns with practical business requirements. Eight segments provide sufficient granularity for differentiated marketing strategies, enabling targeted campaigns for distinct demographic profiles.

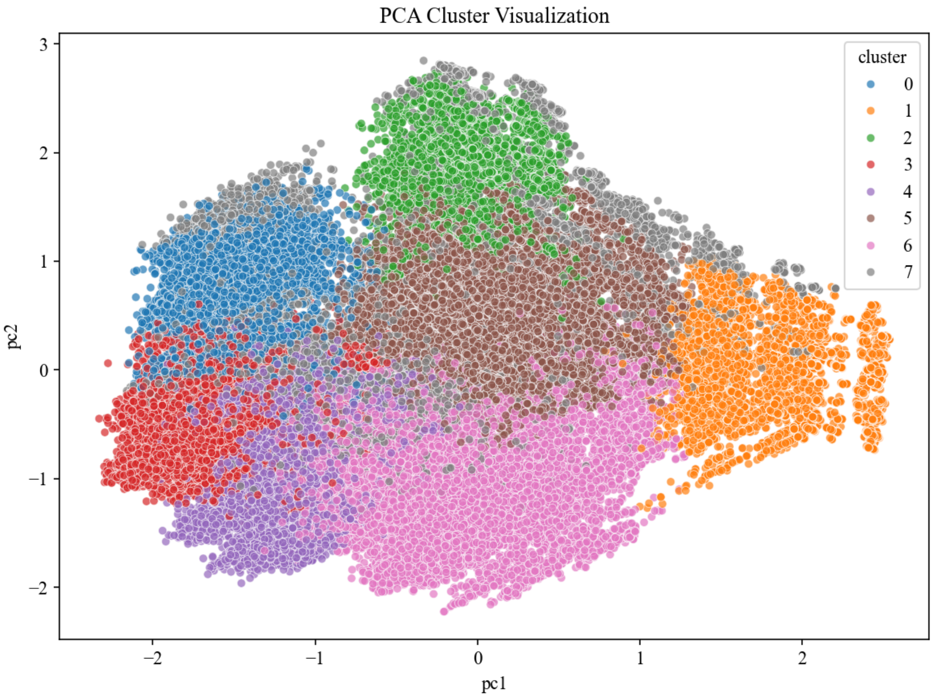


Figure 4: PCA 2-dimensional cluster visualizes the eight clusters projected onto the first two principal components. While some overlap exists in this reduced 2D representation, distinct cluster regions are identifiable, indicating meaningful separation in the original high-dimensional feature space.

3.4 Segment Analysis

3.4.1 Categorical : Lift

For categorical features, I compute lift, which measures how over or under represented a category is within a cluster relative to the overall population.

$$\text{Lift} = \frac{p_{\text{cluster}}}{p_{\text{overall}}} \tag{3}$$

where p_{cluster} is the proportion within the cluster and p_{overall} is the proportion across the dataset.

A lift greater than 1 indicates the category is over represented in the cluster, while a lift less than 1 indicates under representation. For example, a lift of 3.0 means cluster members are three times more likely to belong to that category than the general population.

3.4.2 Numeric Features: Effect Size

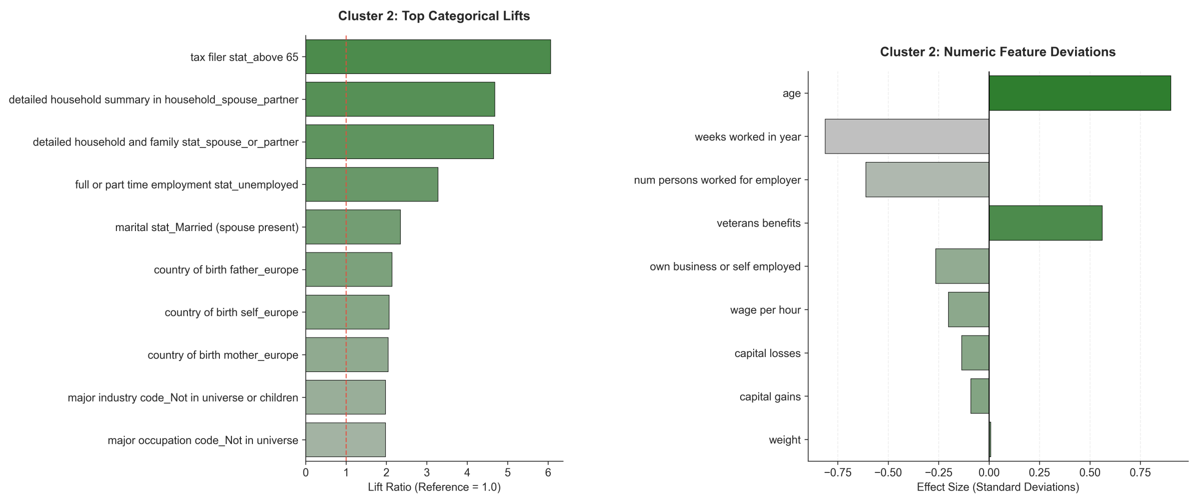
For numeric features, I compute effect size (Cohen's d), which measures the standardized difference between the cluster mean and the global mean.

$$\text{Effect Size} = \frac{\mu_{\text{cluster}} - \mu_{\text{overall}}}{\sigma} \tag{4}$$

where μ_{cluster} is the cluster mean, μ_{overall} is the global mean, and σ is the standard deviation. Since numeric features were standardized prior to clustering, this simplifies to the cluster mean in standard deviation units.

Table 7: Segment profiles and marketing strategies.

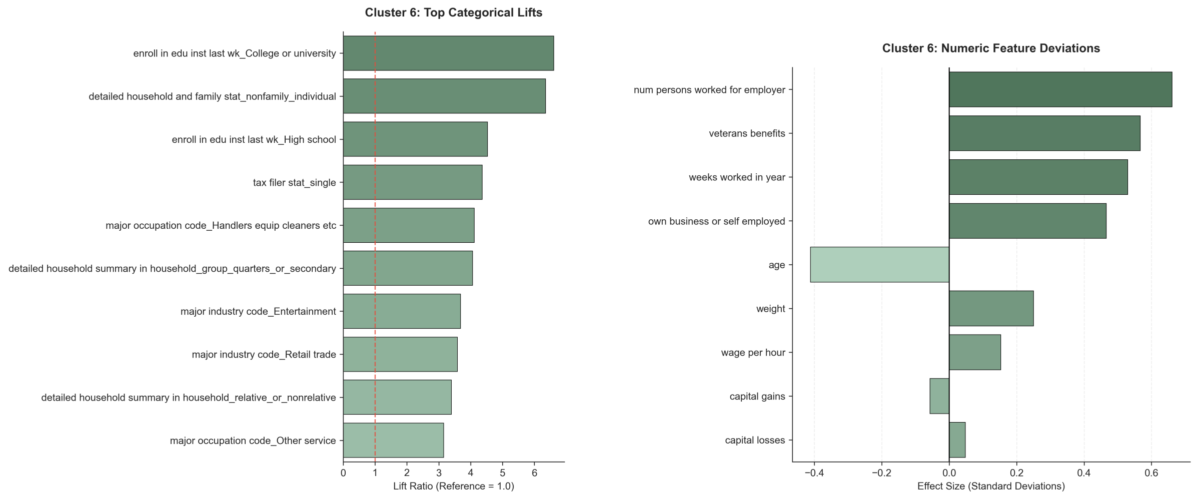
Segment	**Profile**	**Marketing Strategy**
0 - Working Spouses	Spouses of householders employed in education, administration, and medical services; high workforce participation	Household financial products, family insurance bundles, convenience-oriented services
1 - Dependent Minors	Children and non-working young dependents	Not directly targetable. Useful for household composition insights.
2 - Retired Seniors	Previously married; unemployed; age 65+; minimal annual work weeks	Medicare supplements, senior discounts, estate planning, leisure offerings. Prioritize accessibility in communications.
3 - Blue-Collar Affluent	Mining, utilities, and sanitary services workers; income $\geq$\$50K; older demographic; extensive work history across multiple employers	Premium durable goods, retirement planning, home improvement financing, loyalty programs rewarding long-term customers.
4 - Single Professionals	Previously married, currently single; middle age demographic; employed in medical or communications sectors	Individual financial products, travel experiences, health and wellness services. Emphasize independence and self-investment.
5 - Retired Veterans & Seniors	Elderly individuals living in group quarters; previously married; unemployed; low educational attainment;	Medicare supplements, veterans-specific services and discounts, assisted living products. Prioritize accessible communications and emphasize trust and security.
6 - Students & Young Workers	Young individuals enrolled in university or high school; single; high work frequency across multiple part-time employers	Entry-level credit products, student discounts, budget-friendly offerings, brand loyalty programs to capture lifetime value early.
7 - Immigrant Communities	Foreign-born individuals, predominantly from Latin America; diverse racial backgrounds; no distinct income pattern	Culturally tailored marketing, multilingual communications, remittance services, community-focused promotions.



(a) Top categorical lifts

(b) Numeric feature deviations

Figure 5: Cluster 2 - Retired Seniors (High age, less weeks worked, more veterans benefits etc.)



(a) Top categorical lifts

(b) Numeric feature deviations

Figure 6: Cluster 6 profile analysis - Students and Young workers. Marked by lower age, more weeks worked, enrolled in college, etc.

3.5 Segment Descriptions

Table 7 summarizes the demographic profile and recommended marketing strategy for each identified segment.

4 Conclusion

This study applied machine learning to the U.S. Census Income dataset for income classification and population segmentation. The XGBoost classifier achieved an AUC ROC of 0.95, with threshold selection enabling practitioners to balance precision and recall based on business objectives. K Means clustering identified eight demographically interpretable segments, each with distinct characteristics suitable for targeted marketing strategies. While limitations exist including dataset age and K Means assumptions of spherical clusters the combined framework demonstrates how predictive and descriptive analytics can transform demographic data into actionable business intelligence.

References

This work draws on established machine learning methodologies and software tools. The comparison of tree-based ensemble methods versus deep learning on tabular data follows Shwartz-Ziv & Armon (2022), which demonstrated that gradient boosting methods consistently outperform neural networks on structured datasets. The XGBoost algorithm implementation is based on Chen & Guestrin (2016), which introduced the scalable tree boosting framework with regularization. The K-Means clustering algorithm follows the classical formulation by MacQueen (1967), with cluster quality assessed using the silhouette score (Rousseeuw, 1987). Implementation utilized the scikit-learn library for preprocessing, clustering, and evaluation metrics (Pedregosa et al., 2011), and the XGBoost library for gradient boosting classification (XGBoost Developers, 2024). Large language models were used to assist with LaTeX document formatting and Python visualization code generation.